LESSON 4.2
GENERATING EQUIVALENT NUMERICAL EXPRESSIONS

LESSON INTRODUCTION

MA.8.NSO.1.3 Extend previous understanding of the laws of exponents to include integer exponents. Apply the laws of exponents to evaluate numerical expressions and generate equivalent numerical expressions, limited to integer exponents and rational number bases, with procedural fluency.

LEARNING TARGETS

- I can generate equivalent numerical expressions for an expression with a rational base and integer exponent by applying the laws of exponents.

BENCHMARK COHERENCE

BUILDING ON

MA.7.NSO.1.1 Know and apply the laws of exponents to evaluate numerical expressions and generate equivalent numerical expressions, limited to whole-number exponents and rational bases.

WORKING TOWARD

MA.912.NSO.1.1 Extend previous understanding of the laws of exponents to include rational exponents. Apply the laws of exponents to evaluate numerical expressions and generate equivalent numerical expressions involving rational exponents.

ESSENTIAL QUESTIONS

- How do you rewrite a rational base that has a negative exponent as an equivalent exponential expression with a positive exponent?

LESSON NARRATIVE

This lesson extends students' work with the laws of exponents to include rational-number bases. Students explore how the exponent impacts the value of the numerical expressions with fractional bases. Students are initially guided through applying the negative exponent law to expressions with a rational base. Students work with a partner to analyze two different methods for writing equivalent numeric expressions to have the practice and confidence in being able to fluently write equivalent expressions using more than one exponent law independently.

COMPONENT SUMMARY

4.2.1 WARM-UP *(~5 MIN.)*

Select the expression with larger value and justify

4.2.2 GUIDED INSTRUCTION *(15 MIN.)*

Explore patterns in equivalent expressions that have a rational base and exponent

4.2.3 GUIDED PRACTICE *(10-15 MIN.)*

Use exponent laws to write equivalent expressions

4.2.4 COLLABORATION *(10 MIN.)*

Analyze different methods for writing an equivalent expression

4.2.5 YOUR TURN! *(5-10 MIN.)*

Practice in identifying and writing equivalent expressions with rational bases and negative exponents

4.2.6 WRAP-UP *(~5 MIN.)*

Demonstrate mastery of learning target

RESOURCES

GLOSSARY

Key Terms:

- N/A

Additional Terms:

- Negative exponent law

MATERIALS

- (Optional) Highlighters

ADDITIONAL PREPARATION

- N/A

TEACHER TIPS

COMMON MISCONCEPTIONS

- When working with negative fractions, students may incorrectly take the opposite instead of the reciprocal. For example, rewriting $(\frac{-2}{3})^{-5}$ as $(\frac{2}{3})^{5}$.
- Students may not understand why rational bases with positive exponents are equivalent to their reciprocals with negative exponents. For example: $(\frac{2}{3})^{1} = (\frac{3}{2})^{-1}$.

THINGS TO CONSIDER

- Prompt students to write the expanded form to check their application of the laws of exponents.
- Discuss how the patterns recognized in 4.2.2 Guided Instruction connect to multiplying and dividing fractions.

LITERACY AND LANGUAGE ROUTINES

Number Talks

4.2.1 Warm-Up offers an opportunity to incorporate a Number Talk by asking a variety of students to share and justify their answers. Furthermore, throughout 4.2.2 Guided Instruction, there are more opportunities to use this routine to discuss patterns students notice and how the laws of exponents can be applied to numerical expressions with rational bases.

MATHEMATICAL THINKING AND REASONING (MTR) STANDARD

MA.K12.MTR.5.1 Patterns and Structure

In 4.2.2 Guided Instruction, students discover how to apply the laws of exponents to write equivalent expressions for numerical expressions with rational bases. Students identify patterns they notice as the exponent increases and decreases with numerical expressions with rational bases. They consolidate this understanding as they discover how to apply the negative law of exponents.

DIFFERENTIATE INSTRUCTION

4.2.4 Collaboration provides opportunity for auditory learners to verbally discuss different strategies for writing equivalent expressions by applying the laws of exponents to numerical expressions with rational bases.

Edeline's Work	Franco's Work
$\left(\frac{7}{11}\right)^{-5} = \frac{7^{-5}}{11^{-5}} = \frac{11^5}{7^5}$	$\left(\frac{7}{11}\right)^{-5} = \left(\frac{11}{7}\right)^5 = \frac{11^5}{7^5}$

4.2.1 WARM-UP: MATH ARGUMENT

DIFFERENTIATE INSTRUCTION

Prompt students to consider a simpler problem and then apply the same logic to this problem. For example: Does 6^3 or 3^6 result in a greater value?

ENRICHMENT

Does this hold for all rational numbers a and b, when ab ^{a}? For example, which expression has the greater value? $(\frac{1}{2})^6$ or $6^{(\frac{1}{2})}$?

4.2.1 Warm-Up: Math Argument

1. Without performing any calculations, circle the expression you think has the greater value.

   60^{63} $\qquad$ 63^{60}

2. Justify why you selected that expression.
   Answers will vary. The exponent of the first expression is larger (63).

4.2.2 Guided Instruction: Rational Number Bases

1. Consider the table shown.

Exponential Expression	Expanded Form	Equivalent Expression	
$\left(\frac{2}{5}\right)^3$	$\frac{2}{5}\cdot\frac{2}{5}\cdot\frac{2}{5}$	$\frac{2^3}{5^3}$	$\div\frac{2}{5}$
$\left(\frac{2}{5}\right)^2$	$\frac{2}{5}\cdot\frac{2}{5}$	$\frac{2^2}{5^2}$	$\div\frac{2}{5}$
$\left(\frac{2}{5}\right)^1$	$\frac{2}{5}$	$\frac{2^1}{5^1}$	$\div\frac{2}{5}$
$\left(\frac{2}{5}\right)^0$	1	$\frac{2^0}{5^0}$	

Complete the statement to describe the pattern in the table.

As the value of the exponent ○ increases ○ decreases by 1 in the column on the left, the equivalent expression in the column on the right can be found by ○ multiplying ○ dividing the previous value by the ○ base. ○ exponent.

Continuing the pattern, the next row in the table would begin with the expression $\left(\frac{2}{5}\right)^{-1}$.

To continue the pattern, find the value of $1 \div \frac{2}{5}$. Since dividing by a fraction is equivalent to multiplying by the fraction's reciprocal, the result it as follows.

$$1 \div \frac{2}{5} = 1 \cdot \frac{5}{2} = \frac{5}{2}$$

Since $\left(\frac{2}{5}\right)^{-1} = \frac{5}{2}$ and the identity exponent law states $\frac{5}{2} = \left(\frac{5}{2}\right)^1$, it can be concluded that $\left(\frac{2}{5}\right)^{-1} = \left(\frac{5}{2}\right)^1$.

2. Continue the pattern based on this information.

Exponent	Expanded Form	Equivalent Expression	
$\left(\frac{2}{5}\right)^0$	1	$\frac{2^0}{5^0}$	$\times\frac{5}{2}$
$\left(\frac{2}{5}\right)^{-1}$	$\frac{5}{2}$	$\left(\frac{5}{2}\right)^1$	$\times\frac{5}{2}$
$\left(\frac{2}{5}\right)^{-2}$	$\frac{5}{2}\cdot\frac{5}{2}$	$\left(\frac{5}{2}\right)^2$	$\times\frac{5}{2}$
$\left(\frac{2}{5}\right)^{-3}$	$\frac{5}{2}\cdot\frac{5}{2}\cdot\frac{5}{2}$	$\left(\frac{5}{2}\right)^3$	

3. Complete the statement.

A rational base with a negative exponent, $\left(\frac{a}{b}\right)^{-m}$, can be rewritten using the ○ reciprocal ○ opposite of the base with the ○ reciprocal ○ opposite exponent.

4.2.2 GUIDED INSTRUCTION: RATIONAL NUMBER BASES

DIFFERENTIATE INSTRUCTION

Consider describing the expressions verbally to provide an entry point for auditory learners. Also, provide an entry point for visual learners by modeling fraction division and multiplication using number lines or area model.

TEACHER MOVES

Consider discussing with students how the patterns in the tables in this lesson compare to the patterns in the tables in the previous lesson. When you evaluate a rational base to a positive exponent, is the value greater or less than the rational base? How does this connect to the rules for multiplying and dividing fractions?

ENRICHMENT

What is the reciprocal of a fraction $\frac{a}{b}$, where a,b are integers? What is the difference between taking the reciprocal and taking the opposite?

Explain if the expressions $\left(\frac{2}{3}\right)^1, \left(\frac{3}{2}\right)^{-1}$ are equivalent.

4.2.2 GUIDED INSTRUCTION: RATIONAL NUMBER BASES (CONTINUED)

DIFFERENTIATE INSTRUCTION

Consider having students highlight, circle, and/or draw arrows to keep track of the changes they are making as they apply the negative exponent law. Consider having students take turns verbally explaining rewriting the expressions to their partner. The partner should perform the steps explained by their partner to rewrite the expression. Then for the next expression, the partners switch roles.

4. Re-write each expression as an equivalent exponential expression using positive exponents.

   a. $\left(\frac{2}{3}\right)^{-4} = \left(\frac{3}{2}\right)^{4}$

   b. $\left(\frac{3}{-5}\right)^{-9} = \left(\frac{-5}{3}\right)^{9}$

   c. $\left(-\frac{8}{9}\right)^{-2} = \left(-\frac{9}{8}\right)^{2}$

   d. $\left(\frac{21}{4}\right)^{-5} = \left(\frac{4}{21}\right)^{5}$

4.2.3 Guided Practice: Applying Multiple Exponent Laws to Expressions with Rational Bases

1. The table shows equivalent forms of the exponential expression $\left(\frac{4}{5}\right)^4 \cdot \left(\frac{4}{5}\right)^{-3}$.

   a. Complete the table to find the value of $\left(\frac{4}{5}\right)^4 \cdot \left(\frac{4}{5}\right)^{-3}$.

Exponential Expression	Apply Negative Exponent Law	Expanded Form	Value
$\left(\frac{4}{5}\right)^4 \cdot \left(\frac{4}{5}\right)^{-3}$	$\left(\frac{4}{5}\right)^4 \cdot \left(\frac{5}{4}\right)^3$	$\frac{4}{5} \cdot \frac{4}{5} \cdot \frac{4}{5} \cdot \frac{4}{5} \cdot \frac{5}{4} \cdot \frac{5}{4} \cdot \frac{5}{4}$	$\frac{4}{5}$

   b. Explain how you used the expanded form of the expression to determine the value.

   Answers will vary. When multiplying fractions, the reciprocals give you a value of 1. I multiplied every reciprocal set and was left with $\frac{4}{5}$.

Multiple exponent laws can be applied to write equivalent forms of exponential expressions and to evaluate them.

2. For each expression, apply one or more exponent laws to rewrite as an equivalent expression with a rational base and positive exponent. Check off the exponent laws you applied to generate the expression. Show your work.

Exponential Expression	Equivalent Expression with a Rational Base and Positive Exponent	Exponent Laws Applied
$\left(\frac{4}{5}\right)^4 \cdot \left(\frac{4}{5}\right)^{-3}$	$\frac{4}{5}$	☐ negative exponent ☐ power of a power ☐ power of a product ☐ product of powers ☐ quotient of powers
$\left(\frac{11}{3}\right)^{-5} \div \left(\frac{11}{3}\right)^2$	$\left(\frac{3}{11}\right)^7$	☐ negative exponent ☐ power of a power ☐ power of a product ☐ product of powers ☐ quotient of powers
$\left[\left(\frac{3}{4}\right)^2\right]^{-5}$	$\left(\frac{4}{3}\right)^{10}$	☐ negative exponent ☐ power of a power ☐ power of a product ☐ product of powers ☐ quotient of powers
$\left(\frac{2}{3} \cdot \frac{4}{5}\right)^{-4}$	$\left(\frac{15}{8}\right)^4$	☐ negative exponent ☐ power of a power ☐ power of a product ☐ product of powers ☐ quotient of powers

4.2.3 GUIDED PRACTICE: APPLYING MULTIPLE EXPONENT LAWS TO EXPRESSIONS WITH RATIONAL BASES

DIFFERENTIATE INSTRUCTION

Consider prompting students to draw slash marks through the values in the numerators and denominators of the expanded form to show that they equal 1.

ENRICHMENT

What do you notice about expanded form? What happens when you multiply a number by its reciprocal?

DIFFERENTIATE INSTRUCTION

Consider prompting students to continue to write the expanded form of the expressions to aid in determining the equivalent expression.

4.2.4 COLLABORATION: COMPARING STRATEGIES

TEACHER MOVES

During the debrief, ask some students to share which strategy they prefer and why.

4.2.4 Collaboration: Comparing Strategies

1. Edeline and Franco determined the same equivalent expression for $\left(\frac{7}{11}\right)^{-5}$, but they used different strategies.

   Their work is shown in the table.

Edeline's Work	Franco's Work
$\left(\frac{7}{11}\right)^{-5} = \frac{7^{-5}}{11^{-5}} = \frac{11^5}{7^5}$	$\left(\frac{7}{11}\right)^{-5} = \left(\frac{11}{7}\right)^5 = \frac{11^5}{7^5}$

   a. Discuss both students' strategies with your partner.

   b. What steps did Edeline take to find the equivalent expression?
   Edeline applied the power of a quotient law and then the negative exponent law.

   c. What steps did Franco take to find the equivalent expression?
   Franco applied the negative exponent law and then the power of a quotient law.

   d. Whose method do you prefer?
   Answers will vary. I prefer Franco's method because he applied the negative exponent law first, making the exponent positive.

4.2.5 Your Turn!

1. Match each expression in Column A to an equivalent expression in Column B by drawing arrows.

Column A	Column B
$\left(-\frac{7}{8}\right)^{-2}$	$\left(\frac{7}{8}\right)^{2}$
$\left(-\frac{8}{7}\right)^{-2}$	$\left(-\frac{8}{7}\right)^{2}$
$\left(\frac{7}{8}\right)^{-2}$	$\left(\frac{8}{7}\right)^{2}$
$\left(\frac{8}{7}\right)^{-2}$	$\left(-\frac{7}{8}\right)^{2}$

2. Three expressions are shown.
Circle the two expressions that are equivalent.

$\left(\frac{5}{8}\right)^{-2}\cdot\left(\frac{5}{8}\right)^{4}$ $\qquad$ $\dfrac{\left(\frac{5}{8}\right)^{-2}}{\left(\frac{5}{8}\right)^{-4}}$ $\qquad$ $\left(\left(\frac{5}{8}\right)^{-2}\right)^{4}$

3. Generate an equivalent expression using the laws of exponents. Express your answer using positive exponents.
$\left(\frac{9}{10}\cdot\frac{5}{6}\right)^{-6}$

$\left(\frac{45}{60}\right)^{-6} = \left(\frac{60}{45}\right)^{6} = \left(\frac{4}{3}\right)^{6}$

4.2.5 YOUR TURN!

TEACHER MOVES

On question 1, some students may match different expressions than in the key, including cases of one expression from column A to more than one expression from column B. For example, a student might notice that $(\frac{7}{8})^2$ and $(-\frac{7}{8})^2$ are equivalent expressions as well because they have the same value. In cases where students mark different expressions, consider asking students to explain their reasoning to determine whether there is a misconception about exponent laws or if they are recognizing the equivalence of the values.

ENRICHMENT

Why is the third expression in question 2 not equivalent to the first and second expressions? How could you revise the expression to make it equivalent?

4.2.6 Wrap-Up: Ticket Out the Door

1. Rewrite each expression as an equivalent exponential expression with positive exponents.

a. $\left(\frac{-13}{2}\right)^{-3} = \left(\frac{2}{-13}\right)^{3}$

b. $\left(\frac{1}{5}\right)^{-8} = \left(\frac{5}{1}\right)^{8} = 5^{8}$

2. Generate an equivalent expression using the laws of exponents. Express your answer with positive exponents.
$\left(\frac{-5}{4}\right)^{-7}\cdot\left(\frac{-5}{4}\right)^{-3}$

$\left(\frac{4}{-5}\right)^{7}\cdot\left(\frac{4}{-5}\right)^{3} = \left(\frac{4}{-5}\right)^{10}$

4.2.6 WRAP-UP: TICKET OUT THE DOOR

DIFFERENTIATE INSTRUCTION

The information provided from the students here, coupled with what was observed during instruction, can provide information for differentiating instruction in later lessons.

TEACHER MOVES

Proficiency on this 4.2.6 Wrap-Up includes (1) fluency in applying the negative exponent law and (2) fluency in using more than one exponent law to write an equivalent expression for a numeric expression with a rational base and negative exponent.